

EmoSense: AFRN-RAG Framework for FER-Based Facial Emotion Analysis and Mental Health Support

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Abstract. Mental health monitoring is becoming increasingly important with the rising prevalence of psychological disorders, yet traditional assessment methods remain subjective and resource-intensive. This paper presents a hybrid framework that combines an Adaptive Feature Recalibration Network (AFRN) for facial emotion recognition with Retrieval-Augmented Generation (RAG) for mental health-related feedback. The AFRN architecture incorporates two key components: adaptive spectral recalibration modules that enhance emotion-relevant features through learned attention mechanisms, and a progressive multi-scale fusion scheme that captures both fine-grained local patterns and global facial configurations. On top of the FER backbone, the RAG component leverages domain-specific psychological resources to generate contextually informed textual summaries and guidance based on detected emotion trajectories. Experimental validation on a standard facial emotion recognition dataset demonstrates that AFRN achieves improved classification performance compared to a baseline CNN, while maintaining computational efficiency suitable for real-time or near real-time applications. The overall EmoSense framework illustrates a feasible pathway from automated emotion detection to psychologically grounded, human-readable feedback, and is positioned as an exploratory tool for AI-assisted mental health monitoring and clinical decision support rather than a standalone diagnostic or therapeutic system.

Keywords: adaptive feature recalibration, facial emotion recognition, mental health monitoring, clinical decision support, retrieval-augmented generation

1 Introduction

Mental health disorders affect a substantial proportion of the global population [1], and timely support remains a major challenge for healthcare systems. Traditional psychological assessment relies on clinical interviews and self-report questionnaires, which limit accessibility and continuous monitoring [2]. This has motivated interest in computational tools that provide auxiliary, objective indicators of emotional and mental states to support, rather than replace, clinical decision-making.

Facial expression analysis has emerged as a promising non-invasive modality for mental health-related assessment, as visible emotional expressions are linked to underlying affective processes [3]. Facial emotion recognition (FER) can be performed with standard cameras in real time and produces interpretable outputs that align with many clinical and psychological frameworks. However, existing FER systems often struggle with subtle expressions, individual differences, and variations in pose or lighting [4], which is problematic when small facial changes may carry clinically relevant information. Deep learning has advanced FER substantially, with convolutional neural networks (CNNs) achieving strong results on benchmark datasets [5]. Yet conventional CNNs treat all feature channels equally, potentially weakening emotion-specific patterns [6]. Moreover, most existing FER pipelines stop at categorical prediction and do not integrate psychological domain knowledge, limiting their usefulness in clinical contexts where higher-level, context-aware interpretations of emotional dynamics are needed [7].

To address these limitations, this paper proposes EmoSense, a hybrid framework for mental health monitoring and clinical decision support that combines two components. First, we introduce an Adaptive Feature Recalibration Network (AFRN) [8] based architecture with learnable attention mechanisms that emphasize emotion-relevant feature channels while suppressing irrelevant variations. The AFRN module uses a dual-pathway recalibration scheme that captures both global statistics and local saliency, and a progressive multi-scale fusion mechanism to aggregate fine-grained muscle activations and global facial configurations. Second, we integrate a Retrieval-Augmented Generation (RAG) based module [9] to bridge the gap between automated emotion recognition and clinical mental health support. The RAG component generates psychologically informed summaries and guidance intended to contextualize emotional patterns for clinicians and users. By combining AFRN and RAG, EmoSense targets two linked challenges: robust, efficient facial emotion recognition and the translation of low-level emotion outputs into human-readable, knowledge-grounded feedback. The framework is positioned as an exploratory auxiliary tool for psychological assessment and monitoring, not as a standalone diagnostic or therapeutic system.

The main contributions of this work are:

- We propose an AFRN-based network architecture with adaptive feature recalibration and progressive multi-scale fusion for robust, efficient facial emotion recognition in mental health-related scenarios.

- We demonstrate an end-to-end pipeline that integrates FER with knowledge-augmented language models, creating a pathway from visual emotion detection to psychologically grounded textual feedback.
- We present EmoSense as a real-time-capable prototype for mental health monitoring and clinical decision support, and discuss its potential and limitations as an auxiliary component in psychological assessment ⁷.

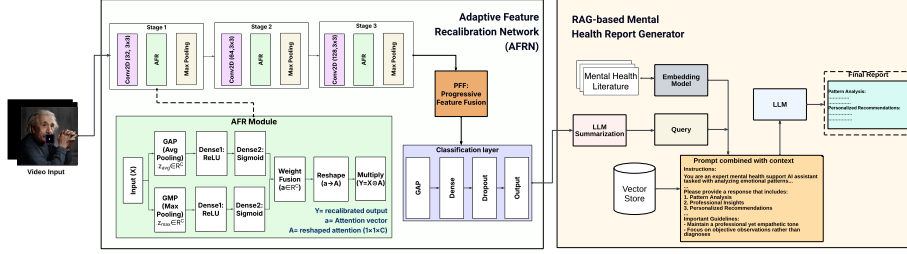


Fig. 1: A high-level representation of the proposed AFRN-RAG framework, showing the integration of an AFRN-based facial emotion recognition module with a Retrieval-Augmented Generation (RAG) system for dynamic mental health analysis.

2 METHODOLOGY

The system architecture in this paper consists of two main components: AFRN-based FER module and RAG-based mental health analysis module. The first component is responsible for generating a series of emotion recognition results based on the user’s dynamic facial expressions, while the second component provides a mental health analysis report based on emotion recognition results. Figure 1 shows an overview of the architecture of the entire system. The details of the AFRN-based architecture and the RAG-based mental health analysis module are in Sections 2.1 and 2.2.

2.1 Adaptive Feature Recalibration Network (AFRN)

The design of AFRN is motivated by several key observations from facial expression analysis. Human facial expressions manifest at multiple spatial scales: subtle muscle movements create fine-grained texture changes, while broader facial configurations form distinct geometric patterns. Furthermore, different facial

⁷ All code for the proposed EmoSense framework is available at <https://github.com/peylix/emosense.git>, offering an open-source implementation to support reproducibility and further research.

regions contribute unequally to emotion recognition-the eye and mouth regions typically carry more discriminative information than other areas [10]. However, traditional CNNs lack mechanisms to adaptively focus on these critical regions and scales.

Drawing inspiration from attention mechanisms in human vision, we hypothesize that a neural network should adaptively determine which features to emphasize based on their relevance to the current classification task. Our AFR module realizes this concept by learning to generate channel-wise attention weights that recalibrate feature importance dynamically [6]. Unlike fixed attention mechanisms, our approach allows the network to adapt its focus based on the specific characteristics of each input image.

Adaptive Feature Recalibration Module The Adaptive Feature Recalibration (AFR) module is the core component of our approach, serving as an intelligent feature selector that enhances emotion-discriminative representations while suppressing irrelevant information. Given an input feature map, the AFR module learns to generate adaptive weights that recalibrate each channel’s contribution to the final representation [11]. The AFR module operates through a dual-pathway design that captures complementary aspects of the input features [12]. The first pathway employs Global Average Pooling (GAP) to extract statistical summary information [13], effectively capturing the overall energy distribution across spatial locations. This pathway is particularly effective in identifying broad patterns and global characteristics that are consistent in different instances of the same emotion [14]. In addition to the statistical summary, the second pathway uses Global Maximum Pooling (GMP) to capture the most prominent responses within each feature channel [15]. This pathway is designed to detect local salient patterns and distinctive features that may be crucial for distinguishing between different emotional states. By combining these two perspectives, our module achieves a more comprehensive understanding of the feature landscape [12].

The mathematical formulation of the AFR module can be expressed as:

$$\mathbf{s}_{gap} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \mathbf{X}_{i,j}, \quad \mathbf{s}_{gmp} = \max_{i,j} \mathbf{X}_{i,j} \quad (1)$$

where $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$ represents the input feature map.

Both pathways are processed through shared fully connected layers to maintain parameter efficiency while generating channel-specific recalibration weights. The shared transformation network consists of two fully connected layers with a bottleneck design that reduces computational overhead while maintaining expressive power. The first layer reduces the channel dimensionality by a factor of 16, followed by ReLU activation, while the second layer restores the original dimensionality and applies sigmoid activation to generate normalized attention weights [16].

The final recalibrated features are obtained by element-wise multiplication between the original features and the fused attention weights from both GAP and

GMP pathways. This design ensures that important features are amplified while less relevant ones are suppressed, leading to more discriminative representations for emotion classification.

Progressive Feature Fusion Strategy Facial expressions exhibit hierarchical patterns requiring multi-scale analysis [17]. Our Progressive Feature Fusion (PFF) mechanism [18] addresses this by intelligently combining features from different network stages, where early stages preserve fine-grained spatial details while deeper stages encode global semantic relationships [19].

PFF first resolves dimensional mismatches using learnable 1×1 convolutions for channel alignment and bilinear upsampling for spatial compatibility. Given features $\mathbf{F}_i$ and $\mathbf{F}_j$ from different stages, the fusion process can be expressed as:

$$\mathbf{F}_{aligned} = \mathbf{F}_i + \text{UpSample}(\text{Conv}_{11}(\mathbf{F}_j)) \quad (2)$$

Rather than fixed weighting, PFF generates adaptive fusion weights based on global context:

$$\mathbf{w}_{fusion} = \sigma(\mathbf{W}_f \cdot \text{GAP}(\mathbf{F}_{aligned}) + \mathbf{b}_f) \quad (3)$$

$$\mathbf{F}_{fused} = \mathbf{F}_{aligned} \odot \text{Reshape}(\mathbf{w}_{fusion}) \quad (4)$$

This adaptive mechanism dynamically adjusts each scale’s contribution based on input characteristics, creating robust multi-scale representations that preserve both local patterns and global structures.

Network Architecture and Training Strategy AFRN consists of three progressive stages with increasing channel capacities ($32 \rightarrow 64 \rightarrow 128$), each incorporating AFR modules. Stage 1 captures low-level features (edges, textures), Stage 2 focuses on facial regions and component relationships, while Stage 3 learns high-level semantic patterns. Following feature extraction, PFF combines Stage 2 and 3 features, creating rich multi-scale representations processed through global average pooling and a 64-unit dense layer with dropout (0.4) for classification.

For robust training, we employ L2 regularization ($\lambda = 0.01$) on convolutional layers and batch normalization after each operation. This lightweight design achieves an optimal balance between computational efficiency and recognition accuracy, which is suitable for real-time applications while maintaining competitive performance on standard benchmarks.

2.2 RAG-based Mental Health Analysis

RAG is a technique designed to enhance the outputs of a Large Language Model (LLM) by integrating external knowledge into the generation process. In typical RAG pipelines, an LLM dynamically retrieves relevant segments from a knowledge base—often stored as vector embeddings—based on semantic similarity to a

query or context [9]. These retrieved segments are then combined with the original query to guide the model’s text generation, thereby improving both the depth and factual accuracy of the final output.

Building upon this principle, our RAG module incorporates a domain-specific external knowledge base containing mental health resources and guidelines. The knowledge base is implemented as a vector store, where documents are systematically segmented into chunks of approximately 1,000 tokens to facilitate efficient retrieval [20]. For the embedding process, we employ Google Gemini’s `text-embedding-004` model, producing high-dimensional vector representations that capture semantic nuances of the text.

To retrieve the most relevant knowledge segments from the vector store, we use cosine similarity between two embedded vectors v_1 and v_2 [21]:

$$sim(v_1, v_2) = \frac{v_1 \cdot v_2}{\|v_1\| \|v_2\|} \quad (5)$$

where $\cdot$ denotes the dot product and $\|v\|$ is the L2 norm of v . This similarity metric enables the system to efficiently identify document chunks that are semantically aligned with the query or contextual input.

Our methodology processes emotional data through a two-stage pipeline. In the first stage, the AFRN-based facial emotion recognition model generates a temporal sequence of emotional states, each accompanied by corresponding timestamps. We denote a sequence of n emotional states by $E = \{e_1, e_2, \dots, e_n\}$, and their corresponding timestamps by $T = \{t_1, t_2, \dots, t_n\}$. A lightweight LLM then summarizes these outputs, extracting key emotional patterns and trends from the sequence [22]. To leverage external mental health knowledge, the summarized emotional profile undergoes the same embedding procedure as the knowledge base documents. The system then computes similarity scores between the summarized profile and all document segments belonging to the vector store, retrieving the top- k segments (where $k = 5$) via:

$$D = \arg \max_k (sim(E(d), E(summary))), d \in Documents \quad (6)$$

Here, $E(d)$ represents the embedding of a document chunk d , and $E(summary)$ the embedding of the emotional summary. Finally, these retrieved documents are concatenated with the emotional summary and a prompt template to guide the LLM in generating a comprehensive mental health report. This report integrates both the automatically summarized emotional patterns and the retrieved, domain-specific mental health knowledge, aiming to ensure both contextual relevance and potential clinical utility.

3 Experiments and Results

3.1 Dataset

In the experiment section, we use the Extended Cohn-Kanade (CK+) dataset containing 593 sequences from 123 subjects with seven basic emotions. The

dataset consists of 327 validated emotion-labeled sequences, which are divided into training and test sets with a ratio of 4 : 1. Our model training uses Adam optimizer with a learning rate $3e-5$ and 5-fold cross-validation. Data augmentation includes $\pm 5^\circ$ rotation, 20% width/height shifts, shear and zoom variations, and flip operations.

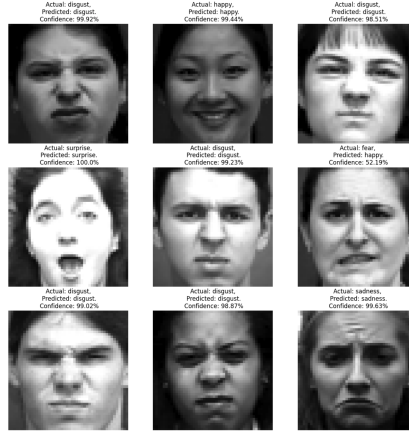


Fig. 2: Examples of AFRN predictions on the CK+ test set. Each panel shows a grayscale face with the ground-truth emotion (Actual), the model’s prediction (Predicted), and the confidence score.

3.2 AFRN-based FER Module

Experimental Details In the experiment section, the framework achieves the capture of real-time frames at $30fps$ and the preliminary detection of the region using the Haar cascade classifier [23]. The preprocessing protocol consists of multiple critical stages: initial frame extraction with Gaussian filtering for noise reduction, followed by comprehensive image standardization. The standardization process includes conversion to grayscale format to preserve the necessary structural information while reducing dimensionality, uniform resizing to 224×224 pixels through bilinear interpolation, and normalization of intensity to the range $[0, 1]$.

Following the preprocessing stage, the network was trained with categorical cross-entropy loss over 200 epochs, with a batch size of 32, with early stopping (patience = 10). To avoid the possibility of an overfitting issue, dropout (with a rate = 0.5) was applied in the classification head.

Experimental Result Analyse The experimental results demonstrated the effectiveness of the AFRN architecture. Figure 3 summarizes the optimization

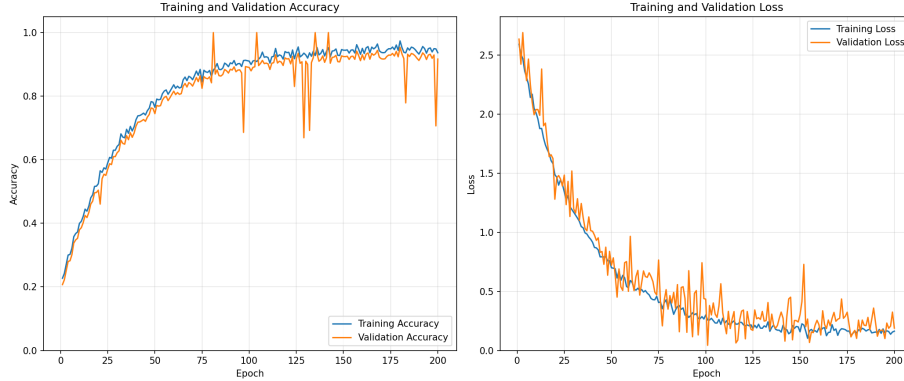


Fig. 3: Training and validation accuracy on CK+ over epochs. Validation accuracy stabilizes near 95.6% while training accuracy plateaus around 94.8%, indicating good convergence and limited overfitting under the chosen protocol.

behavior of AFRN on CK+. The curves show smooth, monotonic improvement with no late-epoch oscillations, indicating stable convergence under the chosen hyperparameters. The validation trajectory closely tracks the training trajectory throughout, suggesting effective regularization and well-matched preprocessing between the two splits. By the end of training, validation accuracy stabilizes around 95.6% while training accuracy plateaus near 94.8%. The small generalization gap, together with the near-parallel trends of train/val curves, supports the claim that the model learns discriminative features without overfitting.

The confusion matrix shown in Figure 4a reveals a strong diagonal, confirming that the majority of samples are classified correctly across all seven emotion categories. Residual errors are concentrated in a few semantically and visually similar pairs. The one-vs-rest ROC curves shown in Figure 4b further quantify robustness at different operating points. Most classes achieve $AUC \geq 0.97$, indicating simultaneously high sensitivity and specificity across thresholds. The micro-averaged ROC also reaches approximately 0.97, consistent with the aggregate accuracy observed in Figure 3. Classes such as *disgust* exhibit particularly strong separability-aligned with their distinctive facial muscle activations—whereas categories involved in the confusions noted above show slightly shallower but still high-performing ROC profiles. Taken together, the ROC results corroborate the accuracy and confusion-matrix findings, demonstrating that AFRN maintains reliable discrimination across emotions and remains stable under threshold variations. Figure 2 displays representative test samples produced by our AFRN on the CK+ dataset. Each tile shows the grayscale input image, the ground-truth label, the model prediction, and the predicted class confidence. Overall, AFRN yields predominantly correct predictions with high confidence—many examples (e.g., *disgust*, *surprise*, *happy*, *sadness*) exceed 95% predicted probability. A few tiles illustrate lower-confidence or incorrect outputs (for ex-

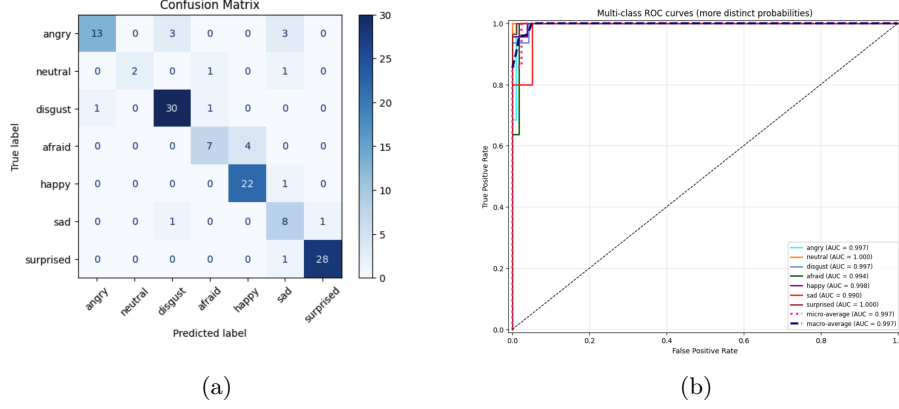


Fig. 4: (a) Confusion matrix on CK+ (rows: true, columns: predicted), showing a strong diagonal and remaining confusions among visually similar classes. (b) Per-class one-vs-rest ROC curves with micro-average; most categories reach AUC ≥ 0.97 , indicating high sensitivity/specificity.

ample, a true fear image predicted as happy at 52%), which typically correspond to low-expression intensity or visually ambiguous poses.

Ablation Study To validate the contribution of each architectural component, we conducted systematic ablation experiments on the CK+ dataset. All configurations were evaluated using identical training protocols (Adam optimizer, learning rate 3×10^{-5} , batch size 32, 5 random seeds). Results are presented in Figure 5. The results reveal three critical findings: (1) Removing dual pooling pathways (GAP/GMP) causes the most severe degradation ($>4\%$), confirming their necessity for capturing complementary global features. (2) Progressive Feature Fusion is the dominant contributor, with only PFF achieving 95.14%, nearly matching the full model. (3) While AFRN’s individual contribution appears modest (1.96% drop), the full model exhibits the lowest standard deviation (0.53), indicating that AFRN primarily enhances training stability rather than peak performance.

Paired t-tests confirm statistical significance ($p < 0.01$) for all ablations except only PFF ($p = 0.082$). The marginal difference between Full and Only PFF suggests potential for architectural simplification in resource-constrained scenarios, though at the cost of reduced robustness. These findings validate that the integration of adaptive recalibration, multi-scale fusion, and dual-pathway aggregation achieves optimal accuracy-stability trade-offs for clinical FER applications.

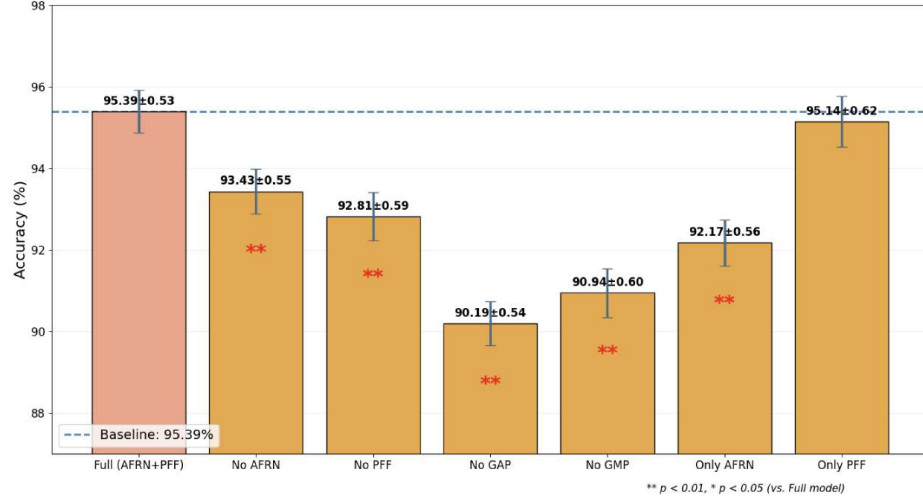


Fig. 5: Ablation study on CK+: mean $\pm$ std validation accuracy for each variant. The full AFRN+PFF performs best; Removing AFR/PFF or GAP/GMP reduces accuracy to varying degrees.

3.3 RAG Module

The implementation of our RAG-based mental health analysis module was built upon the **LangChain** framework, leveraging its comprehensive toolkit for constructing robust retrieval-augmented systems. For the facial emotion recognition (FER) results summarization component, we employed the **gemini-1.5-flash-8b** model, a lightweight large language model selected for its efficiency in processing temporal emotional patterns while maintaining adequate analytical capabilities. The core RAG pipeline utilizes the **gemini-1.5-flash** model as its foundation, chosen for its optimal balance between computational speed and performance quality in generating mental health insights. To assess whether hallucination affects the RAG-generated reports, we developed an evaluation procedure that automatically verifies if the generated content distorts the original facial emotion recognition (FER) results. Specifically, we conducted a series of experiments using three mainstream large language models (LLMs): **gpt-4o**, **claude-sonnet-3.5**, and **gemini-flash-1.5**.

In each experiment, we supplied the generated report to the LLMs and asked them to backtrack the original detected emotion. We then compared the inferred emotion with the actual FER output to identify any significant discrepancies. This process was repeated five times per LLM to ensure the robustness of our findings. Additionally, to mitigate any bias introduced by the explicit mention of the primary detected emotion in the report, we masked those terms during evaluation.

4 Conclusion and Future Work

This paper presented EmoSense, a framework that combines an AFRN-based facial emotion recognition (FER) module with a retrieval-augmented generation (RAG) module to support mental health monitoring and clinical decision support. The AFRN component processes dynamic facial expressions and produces emotion trajectories using a layered architecture with adaptive feature recalibration and multi-scale fusion, aiming to balance accuracy and efficiency for near real-time use. The RAG module then generates text-based summaries and guidance grounded in psychological resources to help users and clinicians interpret emotional patterns. Experiments on a standard facial expression dataset indicate that AFRN improves recognition performance over a baseline CNN. A preliminary system-level evaluation suggests that the generated analyses are perceived as relevant and may increase users’ awareness of their emotional and mental states.

This work is an initial step toward integrating computer vision and knowledge-augmented language models for AI-assisted mental health monitoring. The current study has several limitations: FER is evaluated only on the CK+ dataset with posed expressions, and the RAG outputs have not yet been systematically assessed by mental health professionals. EmoSense should therefore be regarded as an exploratory auxiliary tool rather than a standalone diagnostic or therapeutic system.

In future work, we plan to incorporate richer contextual signals, such as environmental or task-related information, into the FER module and to refine the RAG component with expert-proofread knowledge bases and domain-specific guidelines. We also aim to design a more rigorous evaluation protocol with both end users and clinicians, generating reports for diverse video clips and embedding them in structured questionnaires to assess perceived accuracy, relevance, clarity, and usefulness. These studies will provide more comprehensive validation and inform potential integration of EmoSense into mental health support and telehealth workflows.

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